



PRESIDENCY UNIVERSITY

CSE208: Digital Logic Design Laboratory

Lecture 04: Half Adder and Full Adder

Binary Addition, Adder Design, K-map Derivation and Laboratory Implementation

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Lecture Roadmap

Start from basics

- ▶ Binary addition
- ▶ Sum and carry
- ▶ Half adder

Build the main circuit

- ▶ Carry-in problem
- ▶ Full adder truth table
- ▶ K-map equations

Implement in lab

- ▶ Two half adders + OR
- ▶ IC wiring
- ▶ Verification and errors

Truth Table

Boolean
Expression

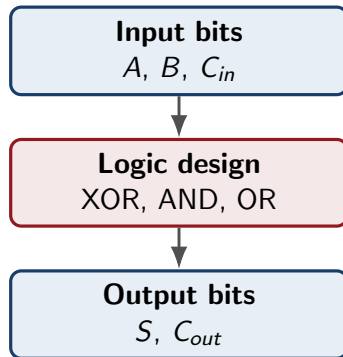
Gate Circuit

Breadboard
Test

Learning Outcomes of This Lecture

After this lecture, students should be able to

- ▶ Explain binary addition using **sum** and **carry**.
- ▶ Design and verify a **half adder** circuit.
- ▶ Explain why a half adder cannot handle a carry-in.
- ▶ Design and verify a **full adder** circuit.
- ▶ Realize a full adder using **two half adders** and **one OR gate**.



Quick Recall: What Is a Combinational Circuit?

Definition

A **combinational circuit** is a digital circuit where the output depends only on the present input values.

Important point

An adder is not a memory circuit. It does not store a previous value; it calculates output instantly from its current inputs.

Design flow used in this lecture



Today's question

How can gates add binary numbers?

Binary Addition: The Smallest Possible Addition

A	B	Binary operation	Sum	Carry
0	0	$0 + 0 = 0$	0	0
0	1	$0 + 1 = 1$	1	0
1	0	$1 + 0 = 1$	1	0
1	1	$1 + 1 = 10_2$	0	1

Sum

the bit written in
the current column

together form addition result

Carry

the bit passed to
the next column

Key idea

When $1 + 1 = 10_2$, the current-column sum is 0 and the carry is 1.

Half Adder: First Addition Circuit

Definition

A **half adder** is a combinational circuit that adds two one-bit binary inputs.

- ▶ Inputs: A , B
- ▶ Outputs: S and C
- ▶ S means **sum**
- ▶ C means **carry**



Main limitation

It has no input for previous carry. That is why we later need a full adder.

Half Adder Truth Table and Equations

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Observe the output pattern

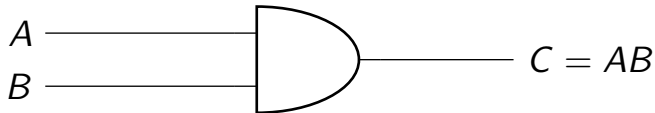
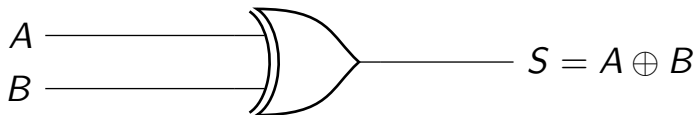
- ▶ $S = 1$ when inputs are different.
- ▶ $C = 1$ only when both inputs are 1.

Therefore

$$S = A \oplus B$$

$$C = A \cdot B$$

Half Adder Gate-Level Circuit



XOR gate

Produces the **sum**: $S = 1$ when the two inputs are different.

AND gate

Produces the **carry**: $C = 1$ only when both inputs are 1.

Why Half Adder Is Not Enough

Problem in multi-bit addition

From the second bit onward, a column may receive a carry from the previous column.

$$\begin{array}{r} 1011 \\ + 0111 \\ \hline 10010 \end{array}$$

Conceptual gap

A half adder can add only:

$$A + B$$

But a real column often needs:

$$A + B + C_{in}$$

Solution

Add a third input named **carry-in**. This gives the **full adder**.

Full Adder: Addition with Carry-In

Definition

A **full adder** is a combinational circuit that adds three one-bit inputs: two data bits and one carry-in bit.

- ▶ Inputs: A , B , C_{in}
- ▶ Outputs: S , C_{out}
- ▶ Used as the building block of multi-bit adders



Remember

C_{in} enters the current stage; C_{out} goes to the next stage.

Full Adder Truth Table

A	B	C_{in}	S	C_{out}	Meaning
0	0	0	0	0	$0 + 0 + 0 = 0$
0	0	1	1	0	$0 + 0 + 1 = 1$
0	1	0	1	0	$0 + 1 + 0 = 1$
0	1	1	0	1	$0 + 1 + 1 = 10_2$
1	0	0	1	0	$1 + 0 + 0 = 1$
1	0	1	0	1	$1 + 0 + 1 = 10_2$
1	1	0	0	1	$1 + 1 + 0 = 10_2$
1	1	1	1	1	$1 + 1 + 1 = 11_2$

Reading the table

The pair $C_{out}S$ represents the binary result of $A + B + C_{in}$.

Deriving the Full Adder Sum

Rows where $S = 1$

001, 010, 100, 111

This means the sum is 1 when the number of input 1s is odd.

Sum equation

$$S = A \oplus B \oplus C_{in}$$

	BC_{in}			
	00	01	11	10
0	0	1	0	1
1	1	0	1	0

K-map pattern confirms XOR behavior.

Deriving the Full Adder Carry-Out

Rows where $C_{out} = 1$

011, 101, 110, 111

Carry is generated when at least two inputs are 1.

Carry equation

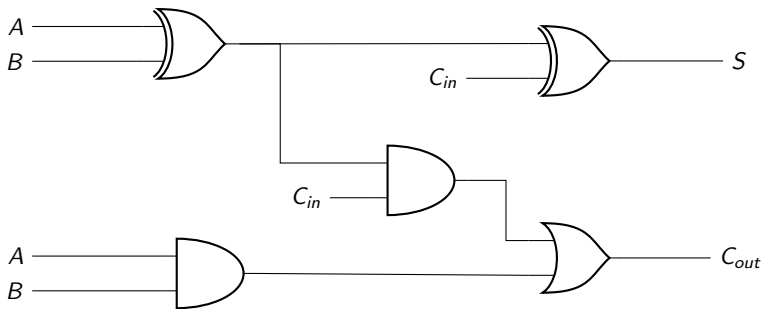
$$C_{out} = AB + AC_{in} + BC_{in}$$

$$C_{out} = AB + C_{in}(A \oplus B)$$

	BC_{in}			
	00	01	11	10
0	0	0	1	0
1	0	1	1	1

K-map groups produce the three product terms.

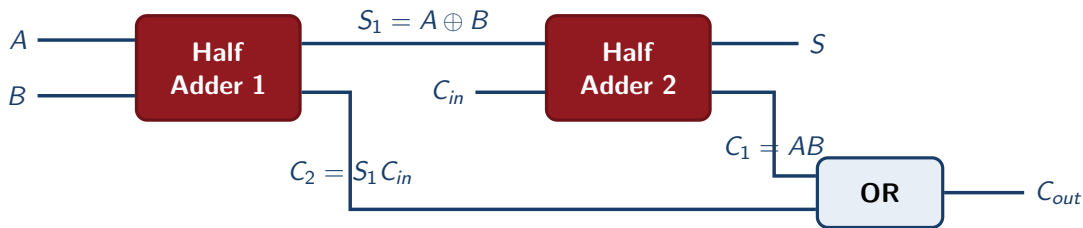
Full Adder Gate-Level Circuit



Circuit equations

$$S = A \oplus B \oplus C_{in}, \quad C_{out} = AB + C_{in}(A \oplus B)$$

Full Adder Using Two Half Adders and One OR Gate



Step 1

$$A + B \rightarrow S_1, C_1$$

Step 2

$$S_1 + C_{in} \rightarrow S, C_2$$

Step 3

$$C_{out} = C_1 + C_2$$

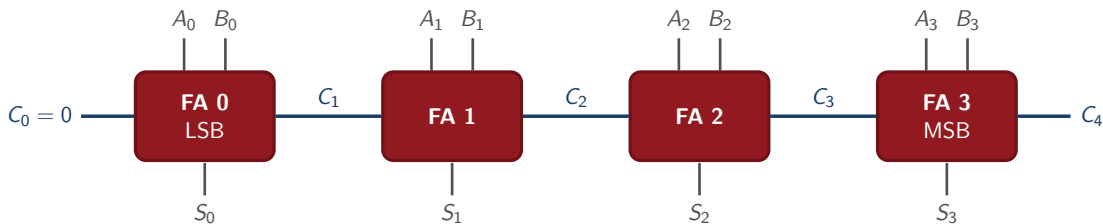
Half Adder vs Full Adder

Feature	Half Adder	Full Adder
Inputs	A, B	A, B, C_{in}
Outputs	S, C	S, C_{out}
Carry-in accepted?	No	Yes
Basic equations	$S = A \oplus B, C = AB$	$S = A \oplus B \oplus C_{in}$
Carry equation	$C = AB$	$C_{out} = AB + AC_{in} + BC_{in}$
Main use	Adds two bits	Builds multi-bit adders

Core difference

The full adder solves the missing carry-in problem of the half adder.

Connecting Full Adders for Multi-Bit Addition



Ripple carry idea

Each stage passes its carry-out to the next stage as carry-in.

Worked Example: One Full Adder Calculation

Given inputs

$$A = 1, \quad B = 0, \quad C_{in} = 1$$

Binary addition

$$1 + 0 + 1 = 10_2$$

So,

$$S = 0, \quad C_{out} = 1$$

Check with equations

$$S = A \oplus B \oplus C_{in}$$

$$S = 1 \oplus 0 \oplus 1 = 0$$

$$C_{out} = AB + AC_{in} + BC_{in}$$

$$C_{out} = 0 + 1 + 0 = 1$$

Laboratory Part A: Half Adder Implementation

Hardware steps

- 1 Identify XOR and AND gate ICs.
- 2 Connect V_{CC} and GND correctly.
- 3 Connect A and B to both gates.
- 4 Connect XOR output to Sum LED.
- 5 Connect AND output to Carry LED.
- 6 Test all four input combinations.

Observation table

A	B	S	C
0	0		
0	1		
1	0		
1	1		

Verification rule

Experimental outputs must match the theoretical truth table.

Laboratory Part B: Full Adder Implementation

Hardware steps

- 1 Build Half Adder 1 using A and B .
- 2 Feed S_1 into Half Adder 2.
- 3 Connect C_{in} to Half Adder 2.
- 4 OR the two carry outputs.
- 5 Test all eight input combinations.

Observation table

A	B	C_{in}	S	C_{out}
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Common Wiring and Testing Mistakes

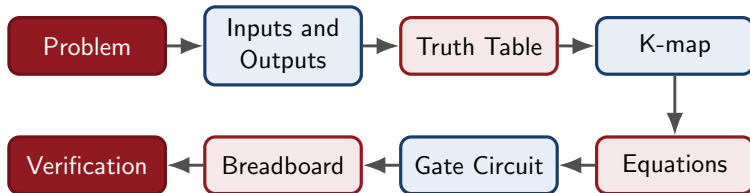
Circuit errors

- ▶ V_{CC} or GND not connected
- ▶ IC inserted in the wrong direction
- ▶ Floating inputs left unconnected
- ▶ LED polarity reversed
- ▶ Output pin connected to another output pin

Conceptual errors

- ▶ Confusing C_{in} with C_{out}
- ▶ Treating $1 + 1$ as decimal 2, not binary 10_2
- ▶ Testing only one or two input cases
- ▶ Forgetting that full adder has eight input cases
- ▶ Mixing half-adder and full-adder equations

Complete Design Chain



What a perfect lab answer should show

Truth table, equations, circuit diagram, connection procedure, observation table and conclusion.

Practice Tasks

Task 1: Half Adder

Design and draw a half adder circuit. Then fill the truth table and write the equations for S and C .

Task 2: Full Adder

Using the full adder truth table, prove that:

$$C_{out} = AB + AC_{in} + BC_{in}$$

Task 3: Circuit construction

Draw a full adder using two half adders and one OR gate. Label A , B , C_{in} , S_1 , C_1 , C_2 , S , and C_{out} .

Task 4: Quick calculation

Find S and C_{out} when $A = 1$, $B = 1$, $C_{in} = 1$.

Summary

Half adder

- ▶ Adds two bits: A and B
- ▶ $S = A \oplus B$
- ▶ $C = AB$
- ▶ No carry-in input

Full adder

- ▶ Adds three bits: A , B , C_{in}
- ▶ $S = A \oplus B \oplus C_{in}$
- ▶ $C_{out} = AB + AC_{in} + BC_{in}$
- ▶ Built using two half adders and one OR gate

Final message

Full adder is the basic building block of practical binary addition.

Thank You

Questions?

Next Lecture: **Half Subtractor and Full Subtractor**